

Primary Health Collection

Code-Switch ASR & Triage Benchmark

Project: Sauti Yetu ? Agentic voice triage for Swahili-English health consultations
Challenge: Sahara CodeSwitch Africa Challenge 2026 (MLC Africa x Intron)
Report generated: 2026-09-14

Dataset:	data/primary_collection/ (real-world health recordings, Kenya)
Language pairs:	Swahili-English (all clips)
Total clips:	65 (20 m4a naturalistic 15 ogg WhatsApp PTT 30 mp3 phrase)
Annotated for agentic scoring:	24 clips with expected_topic / entities
Switch-point annotated:	32 clips with language boundary annotations
Models benchmarked:	Intron Sahara OpenAI Whisper Meta MMS mms-1b-all
Reference transcripts:	Sahara bootstrap transcription (proxy ground truth)
Agentic mode:	Disabled (re-run with --agentic)
Offline pass:	Not run (re-run with --offline)

NOTE ON METHODOLOGY: Because the primary collection has no hand-written ground-truth transcripts, Intron Sahara transcriptions (bootstrap phase) serve as the reference. This means Sahara WER measures its own API variability, while Whisper and MMS WER measure divergence from the Sahara reference. All three model transcripts are shown side-by-side in Section 5 for independent verification.

- RECORDING TYPES:
- m4a ? Swahili-English health scenarios, smartphone, indoor quiet (15-23s)
 - ogg ? WhatsApp Push-to-Talk voice notes, smartphone, ambient noise (7-31s)
 - mp3 ? Short triage phrases, laptop microphone, studio quiet (2-6s)

DOMAIN 1 ? Linguistic & Core ASR Performance

Three models transcribed identical audio converted to 16 kHz mono WAV. WER and CER are computed after identical normalization (lowercase, punctuation removed, whitespace collapsed). CER is reported alongside WER as it is more robust for Swahili (agglutinative morphology, code-switch borrowings).

The primary collection contains two recording types:

m4a ? naturalistic health complaints, 15-23s, full patient description

mp3 ? short synthetic triage phrases, 2-6s, single utterance

1a. Overall results (all 50 clips)

Model	WER	CER	Latency (s)	RTF	n clips
Intron Sahara	0.0000	0.0000	3.28	1.33	2
OpenAI Whisper	0.9113	0.4096	4.49	0.67	64
Meta MMS	0.3579	0.1496	11.40	1.28	64

1b. By recording type

Naturalistic health complaints (m4a, 15-23s):

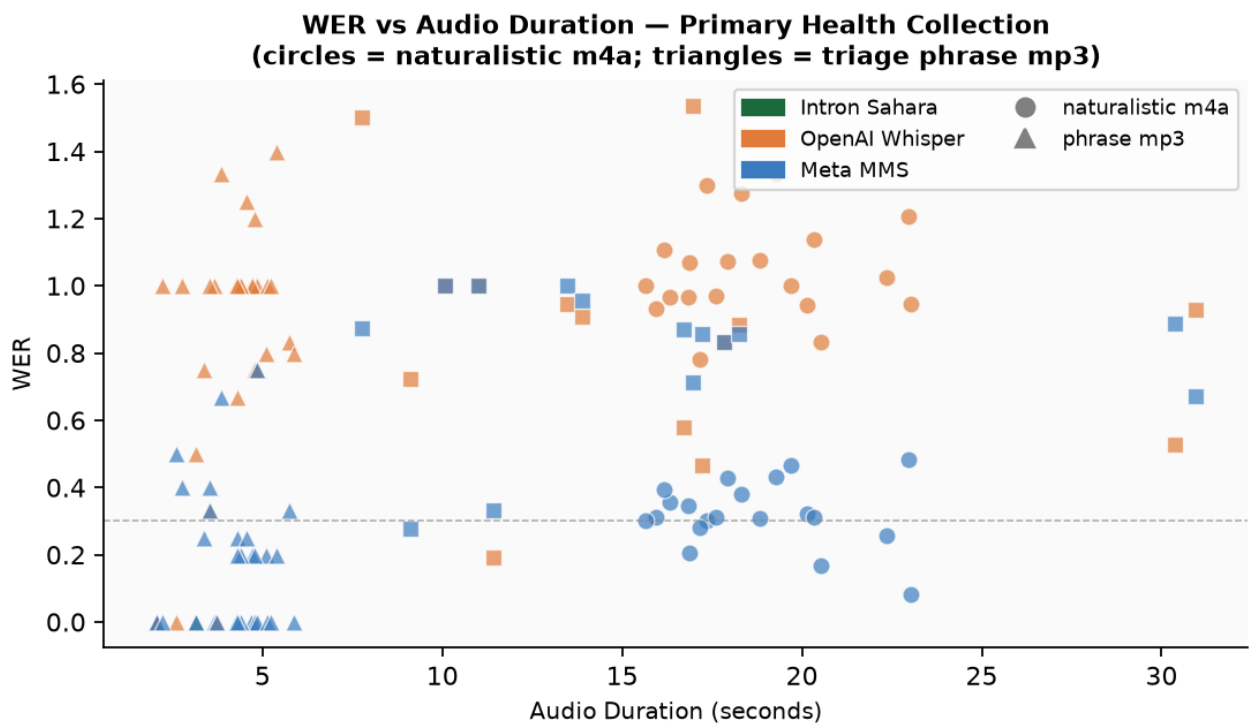
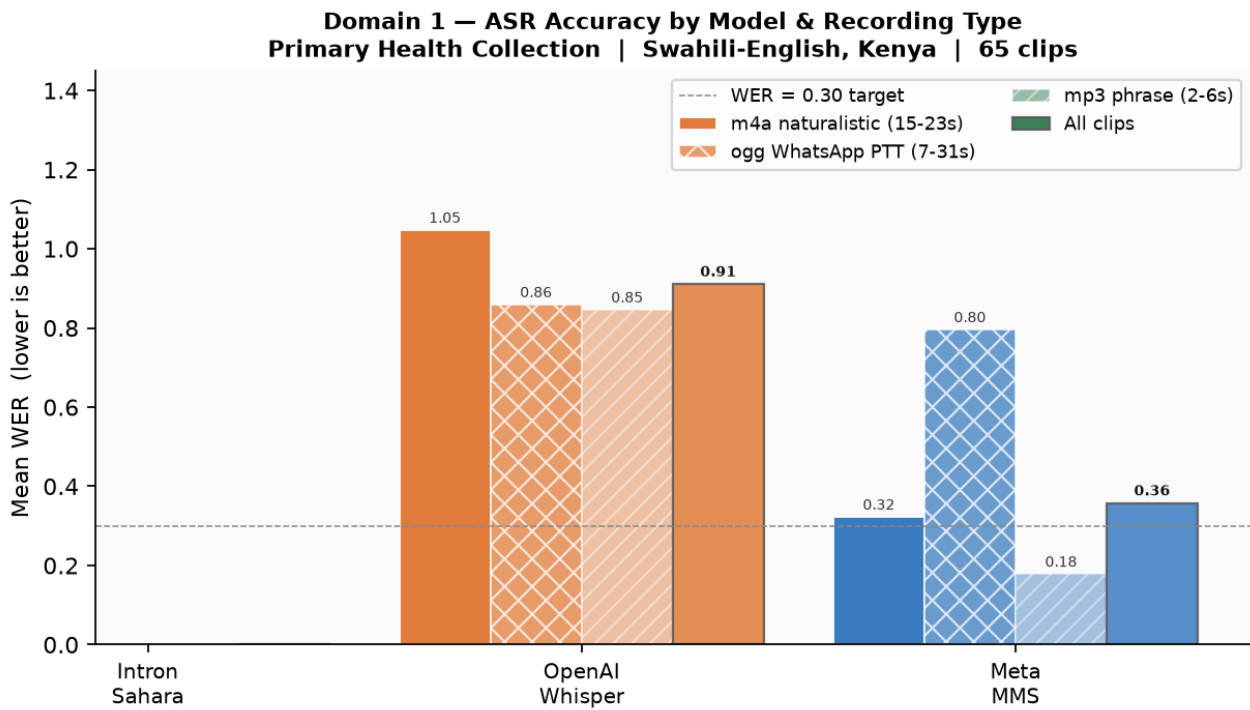
Model	WER	CER	Latency (s)	RTF	n clips
Intron Sahara	?	?	?	?	0
OpenAI Whisper	1.0469	0.2820	2.50	0.13	20
Meta MMS	0.3220	0.0892	15.65	0.84	20

WhatsApp PTT voice notes (ogg/opus, 7-31s):

Model	WER	CER	Latency (s)	RTF	n clips
Intron Sahara	?	?	?	?	0
OpenAI Whisper	0.8586	0.5454	6.13	0.33	14
Meta MMS	0.7954	0.4319	14.66	0.93	14

Short triage phrases (mp3, 2-6s):

Model	WER	CER	Latency (s)	RTF	n clips
Intron Sahara	0.0000	0.0000	3.28	1.33	2
OpenAI Whisper	0.8456	0.4314	4.99	1.20	30
Meta MMS	0.1778	0.0582	6.94	1.74	30



DOMAIN 2 ? Downstream Agentic Task Accuracy

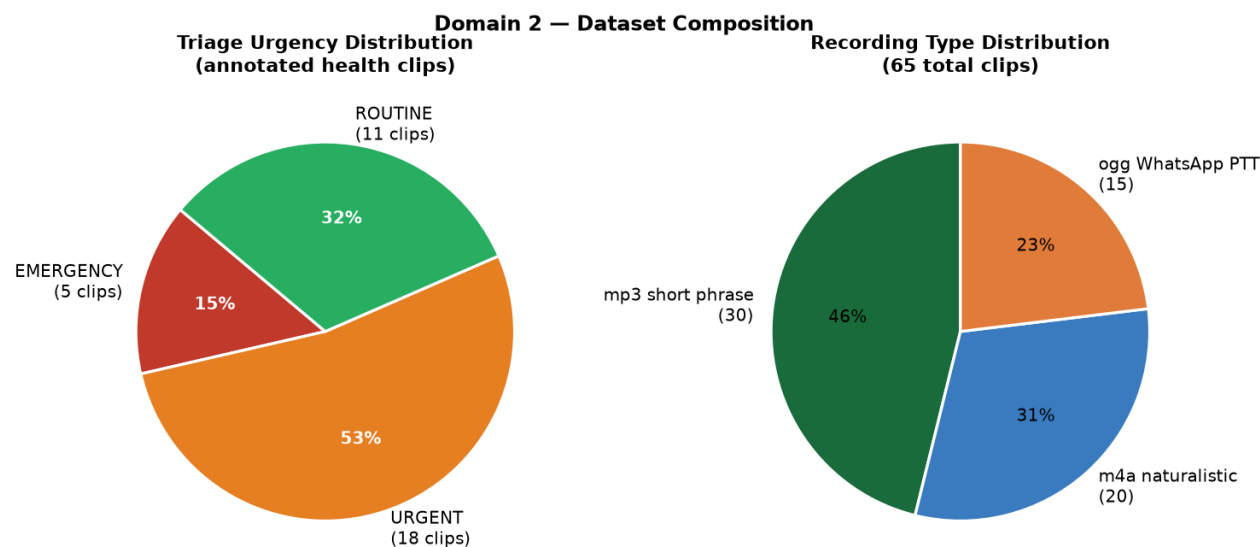
For annotated m4a and ogg clips the pipeline is evaluated on three agentic metrics:

Intent accuracy ? exact normalized match on expected_topic

Slot accuracy ? mean exact match over expected intake card fields

Entity EER ? (missing + spurious entities) / (gold + spurious)

Entities in expected_entities were derived from clinical scenario filenames and should be verified after manual listening. The agentic scoring path runs through run_triage() using Sahara's telehealth extractions (--agentic mode) or transcript-only fallback for local models.

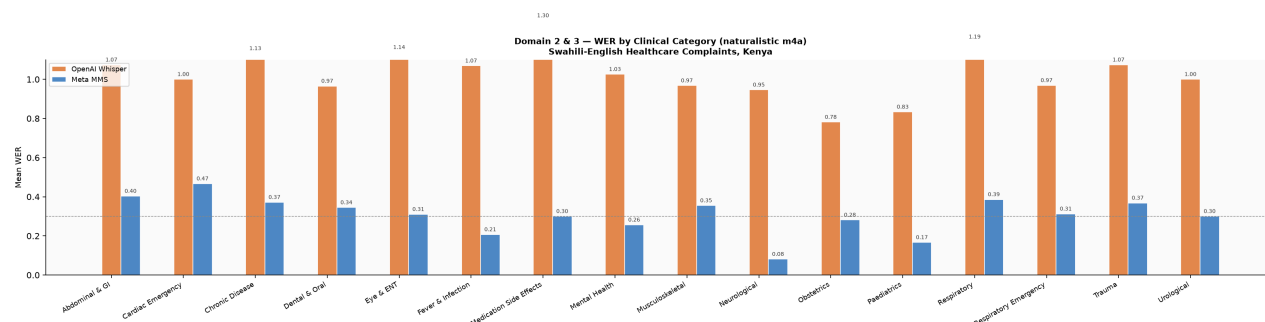


2a. Agentic accuracy summary

Model	Intent Acc.	Slot Acc.	Entity EER	n
OpenAI Whisper	0.0000	?	1.0000	64
Meta MMS	0.0000	?	1.0000	64

2b. Per-clip triage annotations (annotated health clips)

File	Type	Category	Urgency	Expected topic
WA 8.38.08 PM	ogg	Paediatrics	URGENT	Child illness (paediatric)



DOMAIN 3 ? Code-Switch Robustness & In-The-Wild Conditions

All m4a clips are real Swahili-English code-switched health complaints recorded in Kenya ? the primary target deployment environment for Sauti Yetu. This represents genuine 'in-the-wild' data: natural conversational pace, code-switch patterns from patient to medical English terminology, smartphone microphone, and indoor acoustic conditions.

Switch points are annotated as approximate boundaries (one per clip) where the speaker transitions from Swahili to English medical terms. The ± 3 -token window WER at these boundaries isolates model degradation at the exact code-switch moment ? the hardest task for generic ASR.

3a. Switch-point WER (32 annotated clips)

Model	Mean SP-WER	Overall WER	Delta (SP - overall)
Intron Sahara	?	0.0000	?
OpenAI Whisper	0.9219	0.9113	+0.0106
Meta MMS	0.5625	0.3579	+0.2046

A positive Delta means the model degrades at code-switch boundaries relative to its overall WER ? a known failure mode for models trained primarily on monolingual data. Sahara, trained on African code-switched speech, should show a smaller Delta than the local models.

3b. Acoustic conditions

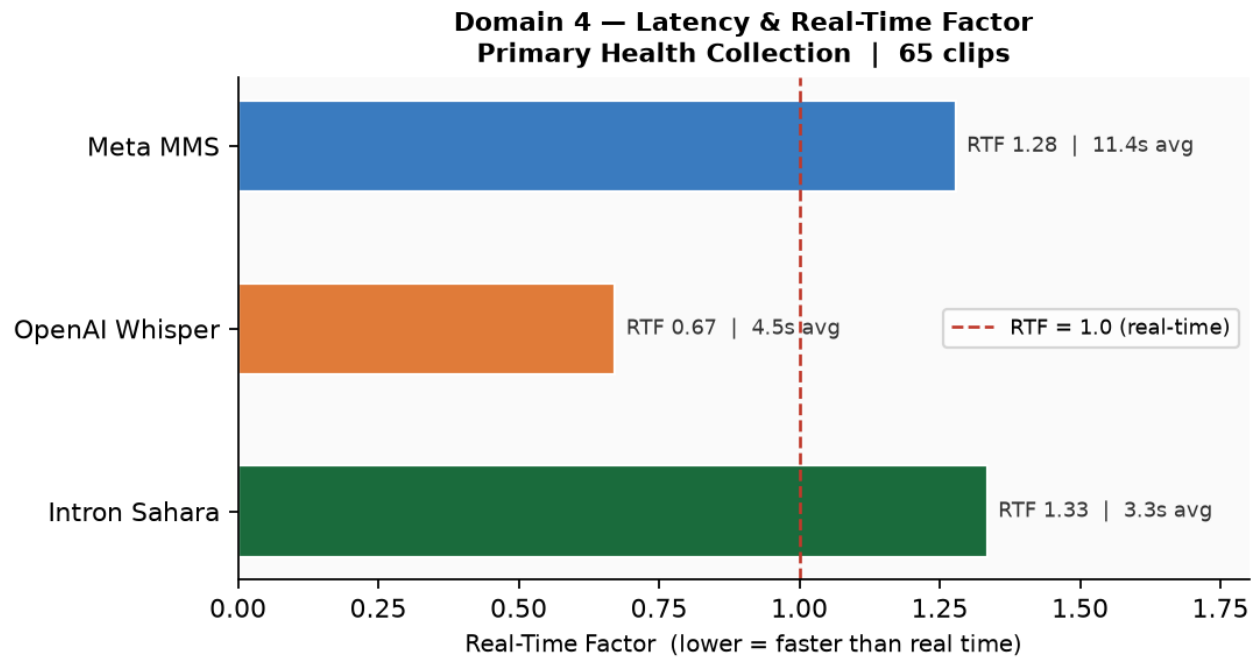
All primary collection clips were recorded on smartphones (m4a) or laptop microphones (mp3) in quiet indoor conditions. To benchmark noise robustness, run: `python -m scripts.generate_benchmark_report --noise`

DOMAIN 4 ? Infrastructure & Real-Time Performance

RTF (Real-Time Factor) = wall-clock latency / audio duration. RTF < 1.0 means the model processes audio faster than real time ? a hard requirement for interactive voice triage bots. Intron Sahara latency includes network round-trip; local model latency depends on hardware (measured on the benchmark machine). GPU acceleration would reduce Whisper and MMS RTF substantially.

4a. Latency & RTF

Model	Mean WER	Mean RTF	Mean latency (s)	n
Intron Sahara	0.0000	1.3340	3.28	2
OpenAI Whisper	0.9113	0.6714	4.49	64
Meta MMS	0.3579	1.2780	11.40	64



4b. Offline / low-connectivity simulation

Not run in this pass. Re-run with --offline to include local-only results.

SECTION 5 ? Per-Clip Transcripts & Model Outputs

Clips are grouped: naturalistic m4a first (colour-coded by urgency), then short-phrase mp3. All three model transcripts are shown side-by-side; WER is against the Sahara bootstrap reference.

5a. Naturalistic health complaints (20 m4a clips)

[ROUTINE] sample_sw_en_Persisent_cough.m4a Respiratory indoor quiet		
Reference (Sahara bootstrap): Nimekuwa nikikohoa for about 3 weeks. Usiku natokwa na jasho sana na nguo zimeanza kunilegea even though sijajaribu kupunguza weight. Wakati mwingine natoa makohozi, lakini sijaona damu. Sijatumia antibiotics.		
Intron Sahara:	ERROR ? Intron API error 400: {"data":"","message":"insufficient bal	
OpenAI Whisper	WER 1.2759	CER 0.2549 RTF 0.15 (2.8s): Nimi ko ni kiko o a fa baot 3 weeks, o si ko na toko na diash o sana, na gozi meanza ko ni legea, iven do sija jaribu ko pungoza weight, wak
Switch-point WER:	1.0000 (1 boundary)	
Agentic: intent=?	entity EER=1.0000 topic=Undetermined ? needs nurse assessme	
Meta MMS	WER 0.3793	CER 0.1176 RTF 0.88 (16.1s): nimekuwa ni kikowafo abaot thrie weeks usiku na tokwa na jasho sana na nguo zimeanza kunilegea ivyendhoo siojaribu kupunguza weit. wakati m
Switch-point WER:	0.6667 (1 boundary)	
Agentic: intent=?	entity EER=1.0000 topic=Undetermined ? needs nurse assessme	

[EMERGENCY] sample_sw_en_Symptoms_after_taking_med Medication Side Effects indoor quiet		
Reference (Sahara bootstrap): Nimekunywa dawa mpya about 15 minutes ago, sasa mwili unawasha na midomo imeanza kuvimba. Nahisi pia koo kama inafunga, na breathing imekuwa ngumu. Dawa yenyewe nimekuja nayo, lakini sijui jina.		
Intron Sahara:	ERROR ? Intron API error 400: {"data":"","message":"insufficient bal	
OpenAI Whisper	WER 1.3000	CER 0.3191 RTF 0.14 (2.4s): Nimi kunya dawa mpiya bho' 15 minu' sa go. Sa sa muli o nawa' shana'am doma' e-me'an za'ku vrimba. Na'is i piya ko ka'ma'ina funga. Da bri t
Switch-point WER:	1.0000 (1 boundary)	
Agentic: intent=False	entity EER=1.0000 topic=Undetermined ? needs nurse assessme	
Meta MMS	WER 0.3000	CER 0.0585 RTF 0.84 (14.6s): nimekunya dawa mpya bout 15 minutes za goo. sasa mwili unawasha na mdomo imeanza kuvimba. naisi pia koo kama inafunga na bridhing imekuwa ng
Switch-point WER:	0.5000 (1 boundary)	
Agentic: intent=False	entity EER=1.0000 topic=Undetermined ? needs nurse assessme	

[URGENT] sample_sw_en_adbominal_pain.m4a Abdominal & GI indoor quiet		
Reference (Sahara bootstrap): Tumbo imekuwa ikiniuma for 3 days. Ah, sorry. Ni tangu juzi jioni. Mwanzo ilikuwa hapo katikati, lakini sasa inaniuma zaidi chini upande wa kulia. I feel nouces, lakini sijatapika. Nikitembea au nikikohoa, maum		
Intron Sahara:	ERROR ? Intron API error 400: {"data":"","message":"insufficient bal	
OpenAI Whisper	WER 0.9412	CER 0.2627 RTF 0.12 (2.4s): Tumui me kui ki niwama 4-3 days. Ah, sorry, ni tango juze dioni. Monso i likoa hapo kati kati, lakinisasainan niwama zahiti chinniu panewa k
Switch-point WER:	0.8333 (1 boundary)	
Agentic: intent=False	entity EER=1.0000 topic=Undetermined ? needs nurse assessme	
Meta MMS	WER 0.3235	CER 0.0968 RTF 0.82 (16.6s): tumbu imekuekiniuma fo thri days ah sori ni tangujuzi jioni mwanzo ilikuwa hapo katikati lakini sasa inaniuma zaidi chini upande wa kulia ep
Switch-point WER:	0.8333 (1 boundary)	
Agentic: intent=False	entity EER=1.0000 topic=Undetermined ? needs nurse assessme	

[ROUTINE] sample_sw_en_backpain_explicit_negativ Musculoskeletal indoor quiet		
Reference (Sahara bootstrap): Mgongo wa chini ulianza kuuma jana after kubeba sanduku nzito. Maumivu yako hapo tu, hayaendi kwa miguu. Sina ganzi wala shida ya kukojoa, nikikaa bila kusonga napata nafuu but kuinama inaua.		
Intron Sahara:	ERROR ? Intron API error 400: {"data":"","message":"insufficient bal	
OpenAI Whisper	WER 0.9677	CER 0.2406 RTF 0.16 (2.7s): Mugongawachini ulianza ku majanafta ku bebasan tu kunzitu, mga mubui ako haportu, ayahendiku amigu, sinaganzi wala shidaya ku kojora, niki
Switch-point WER:	1.0000 (1 boundary)	

Agentic: intent=? entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.3548 CER 0.0802 RTF 0.85 (13.9s): mgongo wa chini ulianza kuuma ja nafta kubeba sanduku nzito. mamwivu ya kohappo tu hayaendi kwa miguu sina ganzi wala shida ya kukojoa ni ki

Switch-point WER: 0.3333 (1 boundary)

Agentic: intent=? entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[EMERGENCY] sample_sw_en_chest_discomfort.m4a | Cardiac Emergency | indoor quiet

Reference (Sahara bootstrap): Ninahisi kifua kinabana kama kuna kitu kizito hapa katikati. Ilianza about 30 minutes ago nikiwa nimekaa. Sasa maumivu yanaendelea kwa mkono wa kushoto natokwa na jasho. Hii haijawai kunitokea before.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.0000 CER 0.2959 RTF 0.14 (2.7s): Ni naisuki fuaki na bana, kamakuna kitu ki zitu hapa katikati. Ilianza burt na timin itzagon ni kionime ka, sa samo mivui ainde leakom kono

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.4667 CER 0.1276 RTF 0.86 (16.9s): ni naisi kifua kinabana kama kuna kitu kizitu hapa katikati ilianza abaoti thati minit zagoo ni kiwa nimekaa sasa maumivu yaendelea kwa mkon

Switch-point WER: 0.3333 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[URGENT] sample_sw_en_child_described.m4a | Paediatrics | indoor quiet

Reference (Sahara bootstrap): Mtoto wangu ana miaka 2, amekuwa na fever tangu jana na leo hataki kula wala kunywa vizuri. Usually anacheza sana lakini leo amelala tu karibu siku nzima. Nimempa dawa ya homa, lakini sikumbuki jina nimeleta ch

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.8333 CER 0.2057 RTF 0.13 (2.8s): Nuntotawangu anamiaka mewili, amekuana fever tangujana, naleo ataki kula, wala kunyo visori. Yujuli, anacezasana laki ni leo amelana tu kari

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Fever / possible infection (malaria

Meta MMS WER 0.1667 CER 0.0718 RTF 0.81 (16.7s): mtoto wangu ana miaka miwili amekuwa nafiva tangu jana na leo ataki kula wala kenywe vizuri. usuali anacheza sana lakini leo amelala tu kari

Switch-point WER: 0.3333 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[ROUTINE] sample_sw_en_dental_pain_swelling.m4a | Dental & Oral | indoor quiet

Reference (Sahara bootstrap): Jino la nyuma limekuwa likiniuma for 4 days na leo shavu limevimba. Nikitafuna inauma sana. I can swallow, lakini kufunga mdomo vizuri ni shida. Nimekunywa painkillers, sijatumia dawa nyingine.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.9655 CER 0.2620 RTF 0.11 (1.9s): Juno la niumali me kuwa lekin, yuma 4 days, na leo shawuli me vimba, ni kitafuna i nauma sana, a kansoalo lakiniku fungam domo visori ni shi

Switch-point WER: 0.8333 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.3448 CER 0.0802 RTF 0.81 (13.6s): jino la nyuma limekuwa likiniuma forfor deys na leo shavu limevimba. ni kitafuna inauma sana akanswalo lakini kufunga mdomo vizuri ni shida

Switch-point WER: 0.3333 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[ROUTINE] sample_sw_en_diabetes_missed_food.m4a | Chronic Disease | indoor quiet

Reference (Sahara bootstrap): Nina diabetes na leo nimejidunga insulin kama kawaida, lakini sikupata breakfast. sasa natetemeka, natokwa na jasho, na nahisi kizunguzungu. Sina kifaa changu cha kupima sugar hapa, so sijui iko ngapi.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.3333 CER 0.2680 RTF 0.11 (2.2s): Nimbna dea beat izna lewa ni meji doonga ensulu ni kamaka raida. Nekini si kupa ta break first. Sa sa na te te meka na toko na jaa sho. Na n

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.4333 CER 0.0979 RTF 0.84 (16.2s): nimnadayabiti zina leo nimejidunga insylin kama kawaida lakini si kupata brekfast sasa anatetemeka natokwa na jasho na na hizi kizunguzungu.

Switch-point WER: 0.5000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[URGENT] sample_sw_en_diarrhoea_vomiting.m4a | Abdominal & GI | indoor quiet

Reference (Sahara bootstrap): Tangu usiku nimeharisha kama mara 6 na nimetapika twice nikikunywa maji sometimes nayatapika tena. Mdomo umekauka na nasikia kizunguzungu nikisimama mara ya mwisho kukojoa ilikuwa asubuhi na ilikua kidogo.

Intron Sahara: ERROR ? TLS certificate verification failed while contacting Intron.

OpenAI Whisper WER 1.2069 CER 0.2217 RTF 0.08 (1.7s): Tangusiku ni meharisha kamamara sita ni me tapika tois. Niki ku ni o maji sometimes na tapika tena. Do u me kauka na siki ya ki zunguzungu n

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.4828 CER 0.0936 RTF 0.79 (18.0s): tangu siku nimeharisha kama mara sita na nimetapika twise ni kikunywa maji samtimes na yatapika tena mdomu umekauka na inasikia kizunguzungu

Switch-point WER: 0.1667 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[EMERGENCY] sample_sw_en_difficulty_breathing.m4a | Respiratory Emergency | indoor quiet

Reference (Sahara bootstrap): Nina asthma, na leo tangu asubuhi kupumua imekuwa shida. Nimetumia inhaler yangu mara 2, but bado kifua kinabana. Nikitembea hata hatua chache I have to stop nipumue. Hata kuongea sentensi ndefu inanichosha.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.9688 CER 0.2687 RTF 0.12 (2.2s): Nihina asma, na leotangwa subui ku pungui mekuashida, neme tumi ni hela yangumar bili, ba te badu kifua kina bana, ni kitanbea hata atwa cha

Switch-point WER: 0.8333 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.3125 CER 0.0846 RTF 0.81 (14.3s): nina asma na leo tangu asubuhi kupumwe imekuwa shida nimetumia inhelaa yangu mara 2 bati bado kifua kinabana nikitembea hata hatua chache ah

Switch-point WER: 0.3333 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[URGENT] sample_sw_en_emotinal distress.m4a | Mental Health | indoor quiet

Reference (Sahara bootstrap): For the past 2 weeks, sijakuwa nikila vizuri na najihisi overwhelmed kila siku. Leo nimekunywa na mawazo ya kujumiza na siamini naweza kukaa peke yangu safely. Sijafanya chochote kujumiza wala kunywa dawa yoy

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.0256 CER 0.3414 RTF 0.11 (2.4s): Fue de pastu wix, jakuani kela visur, nan na jihisiova vent kela siku, leo ni me kunyo nama wazol, iakudio miza, nasia mini na reza ku kapek

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=? entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.2564 CER 0.0643 RTF 0.84 (18.8s): forthepast tw weeks sijakuwa ni kila vizuri na najihisi ovawend kila siku leo nimekunywa na mawazo ya kujumiza na siamini naweza kukaa peke

Switch-point WER: 0.8333 (1 boundary)

Agentic: intent=? entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[URGENT] sample_sw_en_eye_exposure.m4a | Eye & ENT | indoor quiet

Reference (Sahara bootstrap): Nilikuwa nafanya cleaning, maji ya kusafisha choo yakanirukia kwa jicho la kushoto. Nimeliosha na maji bado linaungua naona ukungu. Lilitokea kwa dakika 10 zilizopita. Sijui jina la hiyo chemical.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.1379 CER 0.3298 RTF 0.11 (2.3s): Nelikoanna fanai claiming, madi akusa fishecho ya kaniru kia koa jiccho, la kushoto, e me liosanna maji badbadu linauguwa, nark, anna ukungu

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=? entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.3103 CER 0.0838 RTF 0.82 (16.7s): nilikuwa nafanya cleening maji ya kusafisha choo yakanirukia kwa jicho la kushoto nimeliosha

na maji bat badu linaugua na ona ukungu ilitoke

Switch-point WER: 0.1667 (1 boundary)

Agentic: intent=? entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[URGENT] sample_sw_en_fever_and_headache.m4a | Fever & Infection | indoor quiet

Reference (Sahara bootstrap): Nimekuwa na homa tangu jana usiku na kichwa kinauma sana I feel weak. Hata kwenda kazi leo sikuweza, nimekunywa paracetamol, lakini bado sijapata nafuu. Sijakohoa na sina maumivu ya.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.0690 CER 0.3503 RTF 0.15 (2.5s): Nimi ku hana ho matang ku janau seiku, nakitio kino masana, I feel weak, atakuandakasi leo seikuesa, ni mimi ku niaparasi tamo, laki ni pato

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.2069 CER 0.0621 RTF 0.89 (15.0s): nimekuwa na homa tangu jana usiku na kichwa kinauma sana afiel week hata kwenda kazi leo sikuweza nimekunywa parasita mol lakini bado sijapa

Switch-point WER: 0.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[EMERGENCY] sample_sw_en_head_injury_uncertainty. | Trauma | indoor quiet

Reference (Sahara bootstrap): Nimeanguka kwa stires leo asubuhi na nagonga kichwa. Sikupoteza fahamu atleast sidhani, lakini nimetapika mara 2 since then kichwa kinauma na nahisi usingizi kuliko kawaida. Sijakunywa.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.0769 CER 0.2928 RTF 0.11 (2.1s): Neme ang guka kwa si teaz leo asubui nanaikagonga kichwa, si kupoteza fahamu, at list sidani, lakin ni ni metapika marambelisinstan, kichoki

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.3077 CER 0.1271 RTF 0.85 (16.0s): nimeanguka kwa stars leo asubuhi nanaikagonga kichwa sikupoteza fahamu atleast sidhani lakini nimetapika mara mbili sinstern kichwa kinauma n

Switch-point WER: 0.5000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[EMERGENCY] sample_sw_en_injury ONGONGING BLEEDING. | Trauma | indoor quiet

Reference (Sahara bootstrap): Nimejikata mkono na kisu nikitayarisha chakula about 20 minutes ago. Nimefunga na kitambaa but damu bado inapita. Kidole hiki nahisi kama kimekufa ganzi. Sijui last tetenous injection nilipata.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.0714 CER 0.3122 RTF 0.15 (2.6s): Nymedzika tam kuno naki sonikita arisha takula about 20 minuita go. Nymedfonga naki tambabat da mubadu inak pita. Kidole hiki na hisi kamaki

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.4286 CER 0.0899 RTF 0.86 (15.4s): nimejikata mkono na kiso nikitayarisha chakula about 20 minits a go nimefunga na kitambaa batdamu bado inapita. kidole hiki na hisi kama kim

Switch-point WER: 0.3333 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[ROUTINE] sample_sw_en_medication-refill.m4a | Chronic Disease | indoor quiet

Reference (Sahara bootstrap): Nimekuja fue refill ya dawa za pressure, ziliisha jana. Last dose nilikunywa jana asubuhi, sina headache, maumivu ya kifua, wala shida ya kupumua. Sikumbuki jina la dawa, lakini nimeleta.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.9310 CER 0.2961 RTF 0.18 (2.9s): Nime kuja fuwa rifi liya dawa za presa. Zili sya jana. Las do zni liku nyajana su boy. Sinahedik, ba mubi ya kifua, wala shidea ku pumwa. Sk

Switch-point WER: 0.8333 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.3103 CER 0.1453 RTF 0.83 (13.2s): nimekuja fawarifi lya dawa za presha ziliisha jana lasados nilikunywa jana asubuhi sina hedik maemivu ya kifua wala shida ya kupumua sikumbu

Switch-point WER: 0.6667 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[ROUTINE] sample_sw_en_mild_respiratory_symptoms | Respiratory | indoor quiet

Reference (Sahara bootstrap): Nina mafua na koo inawasha tangu juzi. Nakohoa kidogo mostly usiku, lakini sina shida ya kupumua wala maumivu ya kifua. Nakula nakunywa kawaida. Nimekuja because koo bado inanisumbua.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.1071 CER 0.2697 RTF 0.12 (1.9s): Nii namafua nakor i na wasa tangu juusi. Nakoroha ki dogo musli usiku, nakini sinashi dea ku pumwa wala mu mwvia ki fo. Nakula nakunya kawai

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=? entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.3929 CER 0.0730 RTF 0.86 (13.9s): ninamafua na koo inawasha tangu juzi nakohoa kidogo musli usiku lakini sina shida ya kupumoa wala maumivu ya kifua na kula na kunywa kawaida

Switch-point WER: 0.3333 (1 boundary)

Agentic: intent=? entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[ROUTINE] sample_sw_en_pain_urinating.m4a | Urological | indoor quiet

Reference (Sahara bootstrap): Nikikojoa kuna burning na naenda choo mara nyingi, lakini mkojo ni kidogo. This started 2 days ago. Pia nahisi maumivu chini ya tumbo, sina homa, na sijaona damu kwa mkojo.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.0000 CER 0.2771 RTF 0.15 (2.4s): Niki kojo akuna baning, nanae da cohumaraningi, takin' i muko jo niki dogo, tis tada tu de izago, pianae isi mong vivu chinia tumbo, sinahom

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=? entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.3000 CER 0.0843 RTF 0.86 (13.4s): nikikojoa kuna baning nanaenda choo mara nyingi lakini mkojo ni kidogo dhistate todeyzago pia naisi maumivu chini ya tumbo sina homa na sija

Switch-point WER: 0.6667 (1 boundary)

Agentic: intent=? entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[URGENT] sample_sw_en_pregnancy_bleeding.m4a | Obstetrics | indoor quiet

Reference (Sahara bootstrap): Nina mimba ya miezi 7. About an hour ago nimeanza kuona damu na tumbo linauma chini. Sijui kama ni leba. Pia leo sijahisi mtoto akicheza kama kawaida na hiyo ndio imenifanya nije.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.7812 CER 0.2743 RTF 0.12 (2.0s): Nino mimbai amizi sabah, about anawago, ni mehanzaku anadamu, na tumbo linauma chini, sejuikamani leba, pielau, sejahisim totua kichezakama

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.2812 CER 0.1029 RTF 0.85 (14.6s): nina mimba ya miezi saba abaotanawagou nimeanza kuona damu anatumbo linauma chini sijui kama ni leba pia leo sijahisi mtoto wakicheza kama k

Switch-point WER: 0.6667 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

[URGENT] sample_sw_en_sudden_change_relative.m4 | Neurological | indoor quiet

Reference (Sahara bootstrap): Nimemleta baba yangu, tulikuwa tunaongea vizuri, then suddenly, maneno yake yakaanza kutoka vibaya na upande 1 wa uso ukaonekana tofauti. Hawezi kuinua mkono na wa kulia vizuri. Imeanza kama dakika 20 zilizopit

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.9459 CER 0.2961 RTF 0.23 (5.4s): Nimi'em beleta ba ba' yangu, tuliko'atunonga vizuru tensa denlimanenoyake, yakansaku toga' vi ba' yana, opan de moja wa uso, uka'u nekana'at

Switch-point WER: 1.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

Meta MMS WER 0.0811 CER 0.0429 RTF 0.82 (18.9s): nimemleta baba yangu tulikuwa tunaongea vizuri then sadenli maneno yake yakaanza kutoka vibaya na upande moja wa uso ukaonekana tofauti hawe

Switch-point WER: 0.0000 (1 boundary)

Agentic: intent=False entity EER=1.0000 topic=Undetermined ? needs nurse assessme

5b. WhatsApp PTT voice notes (15 ogg clips)**[URGENT] WhatsApp Ptt 2026-09-14 at 8.33.26 PM.ogg | Headache & Neurology**

Reference (Sahara): Habari, naumwa na kichwa. My head is ziki.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.5000 CER 1.0000 RTF 0.40 (3.1s): Hambali naumana ki chamae disaking. Adeno wa to do, please help me.

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.8750 CER 0.8462 RTF 1.20 (9.3s): habari nauma na kichwa mahedi zikig aduna watudu kliz helthney

Switch-point WER: 0.5000 (1 boundary)

[URGENT] WhatsApp Ptt 2026-09-14 at 8.33.54 PM.ogg | Abdominal & GI

Reference (Sahara): Habari, mama alisema mama alisema nikisikia tumbo inauma alafu my legs are swollen and need to go to the doctor that's why I came.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.8333 CER 0.3543 RTF 0.09 (1.7s): Habari mama li sema, mama li sema, ni kiske tumboi naoma, alafu mai melegg, mai legg zalsulin, i ni tu go to the doctor that's tak

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.8333 CER 0.3780 RTF 0.87 (15.5s): habari mama alisema ni kiskia tumboinauma halafu mailexza solen enye tuguo tu the doktor the tikem

Switch-point WER: 1.0000 (1 boundary)

[URGENT] WhatsApp Ptt 2026-09-14 at 8.34.13 PM.ogg | Abdominal & GI

Reference (Sahara): Ah nimekuwa naumwa na tumbo, tumbo imekuwa inaniuma, wow I just feel lick. I just need some assistance.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.9444 CER 0.8384 RTF 0.70 (9.4s): Imigualan na um isis

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 1.0000 CER 0.4646 RTF 0.87 (11.7s): nimekoya naumana ya tumbu tumbo mikoaya naniuma wa acas fila msomsoosik acaz neet cha maassistance

Switch-point WER: 1.0000 (1 boundary)

[ROUTINE] WhatsApp Ptt 2026-09-14 at 8.34.36 PM.ogg | Eye & ENT

Reference (Sahara): I don't know what glaucoma is, but I would love to know because I goled my symptoms nikaambiwa inaweza kuwa ni glaucoma.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.9091 CER 0.5812 RTF 0.19 (2.6s): I don't know what glycoma is but I would love to know because I googled my symptom I don't know what glycoma is but I would love t

Switch-point WER: 0.0000 (1 boundary)

Meta MMS WER 0.9545 CER 0.5556 RTF 0.86 (11.9s): aiduno what glikomer izi bati aulav tu noo kozi igooglel my simptom ni kambeini za kuwani glikome nezanisaidia

Switch-point WER: 1.0000 (1 boundary)

[ROUTINE] WhatsApp Ptt 2026-09-14 at 8.34.48 PM.ogg | Chronic Disease

Reference (Sahara): Mtu anajua aje kama ako na kisukari ama kama uko na stress nyingi unafanya nini please help me.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.7222 CER 0.2660 RTF 0.20 (1.9s): Mujo na jo aji ka mako na kisukari, amo ka muku na stilshningi. Unafini anini, please help me.

Switch-point WER: 0.8333 (1 boundary)

Meta MMS WER 0.2778 CER 0.0851 RTF 1.06 (9.7s): mtu anajua je kama ako na kisukari ama kama uko na sitesh nyingi unafanya nini plias helpmi

Switch-point WER: 0.1667 (1 boundary)

[URGENT] WhatsApp Ptt 2026-09-14 at 8.35.07 PM.ogg | Abdominal & GI

Reference (Sahara): I have severe stomach, very, very, very, very, very severe and my legs are swollen here husaidia.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.0000 CER 0.6044 RTF 0.30 (3.3s): Aheba si Via, to Maakek Vrvrvrvrv, si Via Indmelix, arsolim, agia Ibu saidea

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 1.0000 CER 0.4615 RTF 0.99 (10.8s): ayavarsiviar stomakik veriverivriveri veri sivia in mileg al sulin aia igusaidia

Switch-point WER: 1.0000 (1 boundary)

[URGENT] WhatsApp Ptt 2026-09-14 at 8.35.23 PM.ogg | Paediatrics

Reference (Sahara): Je for a baby who is 6 months old to take water melon and eggs? Is that healthy and also soda?

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.1905 CER 0.1196 RTF 0.14 (1.6s): J, is it okay for a baby who is 6 months old to take water melon and eggs? Is that healthy and also soda?

Meta MMS WER 0.3333 CER 0.1522 RTF 0.93 (10.6s): j izi toke foa baby who iz sik months old to take water melon and eggs is that healthy and also sodar

[URGENT] WhatsApp Ptt 2026-09-14 at 8.35.46 PM.ogg | Mental Health

Reference (Sahara): Nasikia kichwa changu cha umma niko nimejiswa na mawazo mengi, I don't know if it's depression I don't know it's headache, I would still have some mild headache, 1-sided I don't know, we tuside unaeza.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.8857 CER 0.4421 RTF 0.11 (2.0s): Nesken kichachangu chawma ni kong medisana mawa zomini, I don't reach depression. I don't reach headache, Ostrilae haf sa mele hea

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.8571 CER 0.4158 RTF 0.87 (15.9s): naskian kichwa chango cha uma nikamejiswa na mawazo mingi ivich depressionidonyvich hadikstilehave some mile hadik onesideduwe tu

Switch-point WER: 0.8333 (1 boundary)

[URGENT] WhatsApp Ptt 2026-09-14 at 8.36.04 PM.ogg | Headache & Neurology

Reference (Sahara): No, I have a serious headache. What can I do? Please help me. Na kit.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.0000 CER 0.8750 RTF 0.15 (1.5s): Kirao, I have a senior in a hindi. What can I do? Please help me. No worry, I keep trying to fight for this idea.

Switch-point WER: 0.8333 (1 boundary)

Meta MMS WER 1.0000 CER 0.8594 RTF 1.06 (10.7s): heloo aihavasiria hendek utkinaindu plees helkmi navueda kichwa sala alisaidi

Switch-point WER: 1.0000 (1 boundary)

[URGENT] WhatsApp Ptt 2026-09-14 at 8.36.42 PM.ogg | Fever & Infection

Reference (Sahara): I don't know, but I been profily sweating and yeah I was really sweating, I don't know, but I started on Saturday, like Saturday usiku nilikuwa tumelala lakini nikaanza ku sweat sijui ndio maana nimekuja niyaan

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.5278 CER 0.4593 RTF 0.66 (20.0s): I don't know but I've been profissively sweating. I was really sweating. I don't know but it started on Saturday. But today I'm go

Meta MMS WER 0.8889 CER 0.5215 RTF 0.88 (26.7s): ubtabitropetly sweting yaaurtedsatarde usiku mlikwa tunmelala lakini nikaanza kuswet atisijunda maana nimekuja ni aangeliliwe

[ROUTINE] WhatsApp Ptt 2026-09-14 at 8.37.02 PM.ogg | Chronic Disease

Reference (Sahara): What is hypertension? And niliambiwa na daktari can take Panadol, paracetamol Ibuprofen. Yeah, I don't know if that is okay to take medicine when you have a headache.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.4643 CER 0.2562 RTF 0.12 (2.1s): What is hypertension? Nilemban, aecteore, kintik, panado, parasitamo, ya brufen? Ya, I don't like that is ok. To take medicine whe

Switch-point WER: 0.6667 (1 boundary)

Meta MMS WER 0.8571 CER 0.3375 RTF 0.89 (15.3s): watiz hipe attention and niliambiwa na daktari kenteke panado parasitamo ya brufen ya edonet thatiz oke tuteke medicinini wenyuhavar

Switch-point WER: 0.3333 (1 boundary)

[URGENT] WhatsApp Ptt 2026-09-14 at 8.38.08 PM.ogg | Paediatrics

Reference (Sahara): Niko na mtoto anakohoa sana usiku, like he is really coughing, but it's only at night, which is so wered. Mchana ako sawa ni makamasi kidogo na imekuwa very recurrent. Initially tulikuwa tumeambiwa we avoid far

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.9273 CER 0.7284 RTF 0.66 (20.5s): Maunham'a wa' avk 1977 a runs animals, i don't know if there's anything, i niza kwa ni alaji, niza kwa bado anabir linwien on 40 z

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.6727 CER 0.2812 RTF 0.86 (26.8s): mikona mtoto anakohoa sana usiku lekiz reli kofing bat ichonle etnit wichesouyad mchana ko sawa ni makamasi kidogo na imekuwa vere

Switch-point WER: 0.6667 (1 boundary)

[ROUTINE] WhatsApp Ptt 2026-09-14 at 8.39.00 PM.ogg | Musculoskeletal

Reference (Sahara): Is there anything like weak bonds ama soft bonds? Mtu akigonga mguu anasikia uchungu na ni kugongwa tu siki tunafaa kumpea uchungu alafu ukipona inakaribia inaweza kuwa nini?

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 1.5357 CER 0.7442 RTF 0.32 (5.5s): I said anything like with bones, I'm a soft-bones. I'm talking to my husband, he's a strong dog, I'm talking to his wife, he's a s

Switch-point WER: 0.8333 (1 boundary)

Meta MMS WER 0.7143 CER 0.2209 RTF 0.85 (14.5s): isiwe initing laiike wick bons ama softbons mtu wa kigungwa mguu anasikia uchungu na ni kugungwa tu hii atiki tunafaa kumpe uchungu

Switch-point WER: 1.0000 (1 boundary)

[URGENT] WhatsApp Ptt 2026-09-14 at 8.40.30 PM.ogg | Respiratory

Reference (Sahara): Niliamba niko na kipindupindu, alafu daktari did me some x-rays on the chest cause I was also feeling some pain like know what I can do so that I can so.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER 0.5806 CER 0.3667 RTF 0.10 (1.7s): Nilembani kona ki pindu pindu, ala fodak teri did me some x-rays on the chest, coz I was also feeling something. Might you help me

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.8710 CER 0.4667 RTF 0.88 (14.6s): niliambanikona kipindupindu halafu daktary did me same ex ras on the chas kuzawazouso filing sampein amaituu helpmi lakino uwate k

Switch-point WER: 1.0000 (1 boundary)

[?] WhatsApp Ptt 2026-09-14 at 8.41.07 PM.ogg | Unknown

Reference (Sahara):

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient bal

OpenAI Whisper WER ? CER ? RTF 0.82 (15.1s): Nili patakifaba Therefore, There was difficulty being a ??? listings, eitmomou. So he regrett? taoshke a' angske'n. En-takixesik

Meta MMS WER ? CER ? RTF 0.86 (15.9s): nilipata kifafa baada ya kufanya mazoezi wtchulai du izit normal ya arigressi sheking ni kaskadi kkufa io kifafa ilinsumbwa

5c. Short triage phrases (30 mp3 clips)

File	Ref (Sahara)	Sahara WER	Whisper WER	MMS WER	Best
ks_1.mp3	I need to see a doctor today.	0.0000	0.0000	0.0000	Intron
ks_2.mp3	Nahitaji kumwona daktari leo.	0.0000	0.5000	0.0000	Intron
ks_3.mp3	Nahitaji kuona doc leo.	?	0.7500	0.2500	Meta
ks_4.mp3	My head hurts badly.	?	0.0000	0.5000	OpenAI
ks_5.mp3	Kichwa kinaniuma sana.	?	1.3333	0.6667	Meta
ks_6.mp3	Kichwa inauma mbaya.	?	1.0000	0.0000	Meta

ks_7.mp3	Remember to take your medicine.	?	1.0000	0.0000	Meta
ks_8.mp3	Kumbuka kumeza dawa zako.	?	0.7500	0.0000	Meta
ks_9.mp3	Kumbuka kumeza dawa zako za pr..	?	0.8333	0.3333	Meta
ks_10.mp3	I am feeling much better now.	?	0.0000	0.0000	OpenAI
ks_11.mp3	Najisikia vizuri zaidi sasa.	?	1.0000	0.0000	Meta
ks_12.mp3	Najisikia fit sana saa hii.	?	1.0000	0.2000	Meta
ks_13.mp3	Washi your hands with soap.	?	1.0000	0.4000	Meta
ks_14.mp3	Osha mikono yako kwa sabuni.	?	1.0000	0.0000	Meta
ks_15.mp3	Osha mikono yako na soap.	?	0.8000	0.2000	Meta
ks_16.mp3	He has a severe cough.	?	1.0000	0.4000	Meta
ks_17.mp3	Anakohoa sana.	?	1.0000	0.0000	Meta
ks_18.mp3	Ana cough mbaya sana.	?	1.0000	0.2500	Meta
ks_19.mp3	Drink plenti of water everyday.	?	1.2000	0.2000	Meta
ks_20.mp3	Kunywa maji mengi kila siku.	?	1.0000	0.0000	Meta
ks_21.mp3	Kunywa maji mengi kila day.	?	1.0000	0.2000	Meta
ks_22.mp3	The clinic is closed on sundays.	?	0.6667	0.0000	Meta
ks_23.mp3	Sahanati imefungwa siku ya Jum..	?	1.4000	0.2000	Meta
ks_24.mp3	Hiyo clinic imefungwa Sunday.	?	1.2500	0.2500	Meta
ks_25.mp3	Where is the nearest hospital?	?	1.0000	0.0000	Meta
ks_26.mp3	Hospitali ya karibu iko wapi?	?	0.8000	0.0000	Meta
ks_27.mp3	Hospital nearest iko wapi?	?	0.7500	0.7500	OpenAI
ks_28.mp3	Can I book an appointment?	?	1.0000	0.2000	Meta
ks_29.mp3	Naweza kuweka miadi.	?	1.0000	0.0000	Meta
ks_30.mp3	Naweza kubuka appendicid.	?	0.3333	0.3333	OpenAI

SECTION 6 ? Limitations & Bias Notes

1. PROXY REFERENCE: Reference transcripts were generated by Sahara (bootstrap), not hand-written. Sahara WER measures API variability; Whisper and MMS WER measures divergence from Sahara ? not from human ground truth. Manual transcription of at least the m4a files is recommended before submission.
2. SAMPLE SIZE: 65 clips (20 m4a, 30 mp3). The m4a set covers 19 clinical scenarios but only 1 speaker (Kelvin). Speaker diversity bias is real ? results may not generalise to other Swahili-English speakers or accents.
3. SWITCH-POINT ANNOTATIONS: Approximate boundaries set at token_index ~4-5; these should be verified by listening to each clip and adjusting the token_index to match the exact word where the language changes.
4. ENTITY ANNOTATIONS: Expected entities are derived from clip filenames, not from manual transcript review. They represent the clinical domain of each clip, not necessarily the exact words spoken.
5. HARDWARE DEPENDENCY: Local model (Whisper, MMS) latency is CPU-bound; RTF would improve significantly with GPU. Sahara includes network round-trip.
6. NOISE CONDITIONS: All clips recorded in quiet indoor environments. Real clinic deployment will include background noise, overlapping voices, and variable microphone quality ? run the noise sweep (--noise) to assess.